ELM362 Analog Haberleşme Sistemleri Final Sınavı 8.6.2007

Süre: 2 saat

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1) (30p.) $x(t) = 2Sinc(40\pi t)$ işareti, transfer fonksiyonu $H(\omega)$ olan bir kanaldan iletilmektedir. Çıkış işareti $y(t) = 20Sinc(40\pi t - 200\pi)$ olduğuna göre,

a) $H(\omega)$'yı bulunuz ve genlik, faz spektrumlarını çiziniz.

b) $h(t)$'yi bulunuz.

c) Bu sistem bozulmasız mıdır? Sistemin girişine $x(t) = t.Sinc^2(\pi t)$ işareti uygulanırsa çıkış ne olur?

d) $x(t) = 2Sinc(40\pi t)$ işareti için, çıkışın enerji spektral yoğunluğunu bulunuz.

e) Çıkış işaretinin enerjisini bulunuz.

a) $A_2 \text{sinc}(t_2) \xleftrightarrow{F} A\,\Pi(f_2)$

$x(t) = \underset{A_2=2}{2}\ \text{sinc}(\underset{2}{40\pi t}) \longleftrightarrow \frac{1}{20\pi}\,\Pi\left(\frac{f}{40\pi}\right) = X(f)$

$A_{40\pi} = 2$

$A = \frac{2}{40\pi} = \frac{1}{20\pi}$

$\underset{A_2=20}{20}\ \text{sinc}(\underset{2}{40\pi t}) \longleftrightarrow \frac{1}{2\pi}\,\Pi\left(\frac{f}{40\pi}\right)$

$A = \frac{20}{40\pi} = \frac{1}{2\pi}$

Zamanda Ötelene

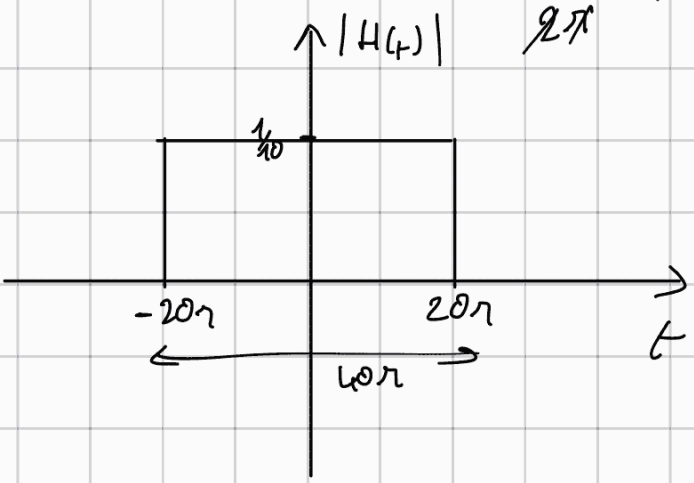
$x(t-t_0) \xleftrightarrow{F} X(f)\,e^{-j2\pi f t_0}$

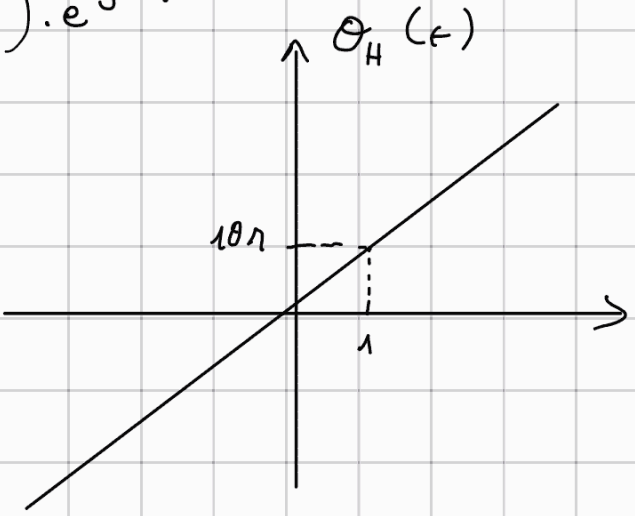
$y(t) = 20\,\text{sinc}(40\pi(t-5)) \longleftrightarrow \frac{1}{2\pi}\,\Pi\left(\frac{f}{40\pi}\right).e^{-j2\pi f 5} = Y(f)$

$x(t) \circledast h(t) = y(t)$

$X(f) \cdot H(f) = Y(f)$

$H(f) = \frac{X(f)}{Y(f)} = \frac{10\,\frac{1}{20\pi}\,\Pi\left(\frac{f}{40\pi}\right)}{\frac{1}{2\pi}\,\Pi\left(\frac{f}{40\pi}\right).e^{-j2\pi f 5}} = \frac{1}{10}\,\Pi\left(\frac{f}{40\pi}\right)e^{j2\pi f 5}$





b) $A\tau\,\mathrm{sinc}(t\tau) \overset{F}{\longleftrightarrow} A \sqcap\!\sqcap (f/2)$

$A = \frac{1}{10}$, $\tau = 40\pi$

$\frac{1}{10} 40\pi\,\mathrm{sinc}(40\pi t) \overset{F}{\longleftrightarrow} \frac{1}{10} \sqcap\!\sqcap (f/40\pi)$

$h(t)$:

$4\pi\,\mathrm{sinc}(40\pi(t+5)) \overset{F}{\longleftrightarrow} \frac{1}{10} \sqcap\!\sqcap (f/40\pi) \cdot e^{j2\pi f 5} = H(f)$

c) Sistem her yerde bozulmasız değildir ama -20π 20π arasında bozulmasız gibi davranır çünkü transfer fonksiyonun genliği sabit açısı lineerdir.

$x(t) = t\,\mathrm{sinc}^2(\pi t)$

zaman ile çarpma

$t^n x(t) \overset{F}{\longleftrightarrow} \left(\frac{j}{2\pi}\right)^n \frac{d^n}{df^n} X(f)$

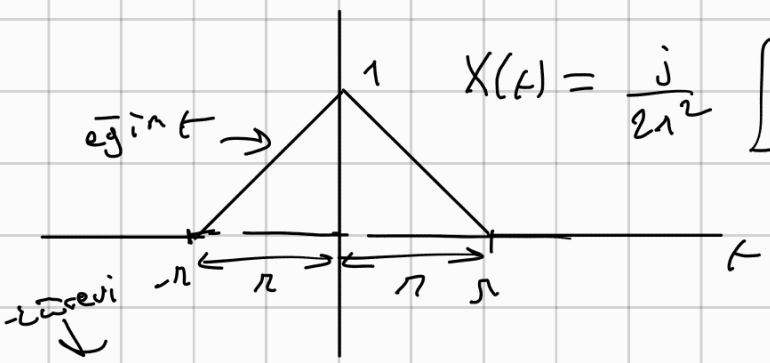
$A\tau\,\mathrm{sinc}^2(t\tau) \overset{F}{\longleftrightarrow} A \wedge (f/\tau)$

$\mathrm{sinc}^2\left(\frac{\pi}{2} t\right) \longleftrightarrow \frac{1}{\pi} \wedge (f/\pi)$

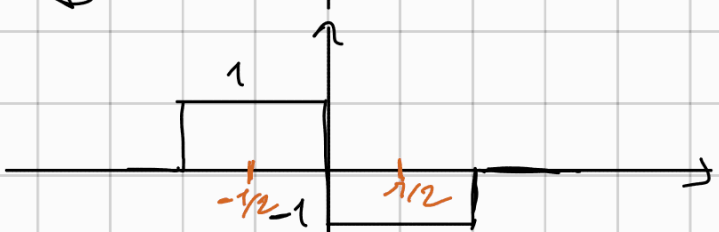
$A\tau = 1$

$A = \frac{1}{\pi}$

$t\,\mathrm{sinc}^2(\pi t) \longleftrightarrow \frac{1}{\pi}\left(\frac{j}{2\pi}\right)\frac{d}{df} \wedge (f/\pi)$



$X(f) = \frac{j}{2\pi^2}\left[\sqcap\!\sqcap\left(\frac{f+\pi/2}{\pi/2}\right) - \sqcap\!\sqcap\left(\frac{f-\pi/2}{\pi/2}\right)\right]$



$$X(f) = \frac{j}{2\pi^2}\left[\Pi\left(\frac{f+\pi/2}{\pi/2}\right) - \Pi\left(\frac{f-\pi/2}{\pi/2}\right)\right]$$

$$Y(f) = X(f)(H_f) \begin{cases} \frac{1}{10}\Pi\left(f/40\pi\right)\cdot e^{j2\pi fs} = H(f) \end{cases}$$

$$Y(f) = \frac{j}{20\pi^2}\left[\Pi\left(\frac{f+\pi/2}{\underbrace{\pi/2}_{2}}\right) - \Pi\left(\frac{f-\pi/2}{\pi/2}\right)\right]\cdot e^{j2\pi fs}$$

$$A_2 \; \text{sinc}(t_2) \xleftrightarrow{F} \underset{\downarrow 1}{A}\,\Pi\left(f/\underset{\downarrow \pi/2}{2}\right)$$
$$\downarrow_1$$

$$\pi/2 \; \text{sinc}\left(\pi t/2\right) \longleftrightarrow \Pi\left(\frac{f}{\pi/2}\right)$$

$$\pi/2 \; \text{sinc}\left(\frac{\pi}{2}(t+s)\right) \longleftrightarrow \Pi\left(\frac{f}{\pi/2}\right)\cdot e^{j2\pi fs}$$

$$\pi/2 \; \text{sinc}\left(\pi/2(t+s)\right).e^{j\pi t\cdot \pi/4} \longleftrightarrow \Pi\left(\frac{f+\pi/2}{\pi/2}\right)e^{j2\pi fs}$$

$$y(t) = \frac{j}{20\pi^2}\left[\pi/2 \; \text{sinc}\left(\pi/2(t+s)\right)e^{j\pi^2 t} - \pi/2 \; \text{sinc}\left(\pi/2(t+s)\right)e^{-j\pi^2 t}\right)$$

$$= \frac{j}{40\pi}\,\text{sinc}\left(\pi/2\,(t+s)\right)\left[e^{j\pi^2 t} - e^{-j\pi^2 t}\right]$$

$$= -\frac{1}{20\pi}\,\text{sinc}\left(\pi/2(t+s)\right)\cdot \underbrace{\frac{e^{j\pi^2 t} - e^{-j\pi^2 t}}{2j}}_{\sin(\pi^2 t)}$$

$$= -\frac{1}{20\pi}\,\text{sinc}\left(\pi/2\,(t+s)\right)\sin(\pi^2 t)$$

d) $x(t) = 2\,\mathrm{sinc}(40\pi t)$ $\quad A_2 = \frac{2}{2} \quad A = \frac{2}{40\pi}$ $\quad 2$

$$\frac{1}{10}\,\Pi\left(\frac{f}{40\pi}\right)\cdot e^{j2\pi f 5} = H(f)$$

$$A_2\,\mathrm{sinc}(t_2) \xleftrightarrow{F} A\,\Pi(f_2)$$

$$X(f) = \frac{1}{20\pi}\,\Pi\left(\frac{f}{40\pi}\right)$$

$$Y(f) = X(f)\,H(f) = \frac{1}{200\pi}\,\Pi\left(\frac{f}{40\pi}\right)\cdot e^{j2\pi f 5}$$

$$\mathcal{E}_g(f) = |X(f)|^2 = \left(\frac{1}{200}\,\Pi\left(\frac{f}{40\pi}\right)\right)^2$$

$$= \frac{1}{40000}\,\Pi\left(\frac{f}{40\pi}\right)$$

e) $$E = \int_{-\infty}^{\infty} \mathcal{E}_g(f)\,df = \int_{-20\pi}^{20\pi} \frac{1}{40000}\,df$$

$$= \frac{40\pi}{40\,000} = \frac{\pi}{1000}\ //$$